

PROFILE

My research centers on **multimodal learning** and **spatial-temporal analysis** for Earth observation, advancing **geospatial reasoning** to deliver actionable insights, especially for **global forest and biodiversity monitoring**.

EDUCATION

Technical University of Munich

Munich, Germany

Ph.D. in Data Science and Earth Observation

2022 - 2026 (*expected*)

- Advisor: Xiaoxiang Zhu. Research focus: Multimodal Learning, Spatiotemporal Analysis, Foundation Models for Remote Sensing

ETH Zurich

Zurich, Switzerland

M.Sc.(Exchange) in Geomatics

2021 - 2022

- Advisor: Konrad Schindler. Focus: Computer Vision, Advanced Signal Processing, Machine Learning for Geospatial Data

KTH Royal Institute of Technology

Stockholm, Sweden

M.Sc. in Geoinformation

2020 - 2022

- Advisor: Yifang Ban. Focus: Deep Learning in Data Science, Computer Vision for Geospatial Applications

Wuhan University

Wuhan, China

B.Eng. in Remote Sensing Science and Technology

2016 - 2020

- Advisor: Shugen Wang. Focus: Digital Image Processing, Remote Sensing Algorithms, Geospatial Analysis

PUBLICATIONS

1. **Yang Mu**, Zhitong Xiong, Yi Wang, Muhammad Shahzad, Franz Essl, Mark van Kleunen, Xiao Xiang Zhu. GlobalGeoTree: A Multi-Granular Vision-Language Dataset for Global Tree Species Classification. 2025. (*Under Review*)
2. **Yang Mu**, Jianhua Guo, Muhammad Shahzad, Xiao Xiang Zhu. National-scale tree species mapping with deep learning reveals forest management insights in Germany. *International Journal of Applied Earth Observation and Geoinformation*, 2025.
3. **Yang Mu**, Muhammad Shahzad, Xiao Xiang Zhu. MPTSNet: Integrating Multiscale Periodic Local Patterns and Global Dependencies for Multivariate Time Series Classification. *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2025.
4. **Yang Mu**, Muhammad Shahzad, Xiao Xiang Zhu. A spectral-spatial-temporal attention network for tree species mapping using DESIS hyperspectral imagery. No. EGU24-10854. *Copernicus Meetings*, 2024.

PROJECTS

Earth Observation Foundation Models for Mine Site Monitoring (MultiMiner)

Horizon Europe

2023.01 - Present

- Developing GSM (Generic Mine Site Monitoring algorithm) with EO foundation model and temporal attention networks to address limited in-situ data
- Applied multi-modal approach (Sentinel-2 time series + EnMAP hyperspectral imagery) to vegetation monitoring tasks
- Achieved significant performance improvements over traditional methods in species richness, leaf area index, and fractional vegetation cover prediction
- Conducted comprehensive evaluation of satellite foundation models (Prithvi, Presto, ALISE, AnySat) for potential integration into the unified monitoring framework

SKILLS	Programming: Python (PyTorch, TensorFlow, scikit-learn), C++, MATLAB, Java	
	Tools: Docker, Git, Linux Shell, HPC/Cloud Computing, Google Earth Engine, QGIS, ArcGIS, GDAL/OGR	
	Languages: Chinese (Native), English (Fluent), German (Intermediate)	
AWARDS AND HONORS	• AAAI Student Scholarship , AAAI Conference, USA	2024.12
	• Erasmus Scholarship , European Commission	2021.09
	• KTH Scholarship (35/1700) , KTH Royal Institute of Technology, Sweden	2020.05
	• Excellence Scholarship , Wuhan University, China	2018 & 2019.10
	• Second Award , Asia and Pacific Mathematical Contest in Modeling, China	2019.02
	• Outstanding Paper , High Resolution Earth Observation Conference, China	2018.10
	• Third place prize , UAV Intelligent Perception Competition, China	2018.08
	• Honorable Mention , Mathematical Contest in Modeling, USA	2018.03
	• Undergraduate Innovation Grant , Wuhan University, China	2017.10
ACADEMIC SERVICES	Reviewers for: <i>Remote Sensing</i> , <i>IEEE Transactions on Geoscience and Remote Sensing</i> .	
	Membership: <i>European Geosciences Union (EGU)</i> , <i>Association for the Advancement of Artificial Intelligence (AAAI)</i> .	